

Drancer AI — Complete Build Roadmap

Personal AI Assistant | Built by Sanjay.K | Robotics & Automation Engineering

Project Overview

Drancer is a locally running personal AI voice assistant built on Ollama + phi3:mini. It runs fully offline on an 8GB RAM Ryzen laptop with no cloud dependency and no API keys. This document covers the complete build roadmap from setup to career portfolio.

Hardware Specs

Component	Spec
RAM	8GB
Storage	256GB
CPU	AMD Ryzen
OS	Windows
AI Model	phi3:mini via Ollama

Prerequisites

- Python 3.10+
- VS Code
- Git
- Ollama (ollama.com)

Phase 1 — Core Brain (Week 1-2)

Goal: Working text chatbot in terminal

- Install Ollama → run: ollama pull phi3:mini
- Create project folder: D:\Drancer with files: brain.py, main.py, voice.py, memory.py, actions.py
- Install Python dependencies: pip install ollama openai-whisper pytsx3 speechrecognition pyaudio
- Build brain.py with Ollama chat integration and Drancer system prompt
- Run python brain.py — confirm Drancer responds

brain.py — Core Code

```
import ollama
SYSTEM_PROMPT = """You are Drancer, a personal AI assistant created by Sanjay.
Your name is DRANCER. Never call yourself anything else. Be concise, direct, and helpful."""
def ask(prompt, history=[]):
    messages = [{"role": "system", "content": SYSTEM_PROMPT}]
    messages += history
    messages.append({"role": "user", "content": prompt})
    response = ollama.chat(model="phi3:mini", messages=messages)
    return response['message']['content']
def test_brain():
    print("Testing Drancer brain...")
    response = ask("Say: Drancer online and ready.")
    print(f"Drancer: {response}")
if __name__ == "__main__":
    test_brain()
```

Phase 2 — Voice (Week 3)

Goal: Speak to Drancer, it responds with voice

- Build voice.py using Whisper tiny model for speech-to-text
 - Add pyttsx3 for text-to-speech output
 - Connect voice pipeline to brain.py
 - Test: speak a question, Drancer replies aloud
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Phase 3 — Memory (Week 4-5)

Goal: Drancer remembers previous conversations after restart

- Build memory.py using JSON storage
 - Store last 20 conversation exchanges
 - Pass conversation history into each Ollama request
 - Test: close terminal, reopen, Drancer recalls past context
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Phase 4 — Actions (Week 6-8)

Goal: Voice commands trigger real PC actions

- Build actions.py: open apps, get time, search web
 - Add intent detection using phi3:mini classification
 - Connect intent to action pipeline in main.py
 - Test: say 'open calculator' — it opens
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Phase 5 — Career Portfolio (Week 9-10)

Goal: Standout project for Robotics/OT/Embedded AI interviews

- Push project to GitHub: github.com/SanjayK-netizen/Drancer-AI
- Write README explaining architecture and setup
- Record a demo video of voice interaction
- Add to resume under Projects section

Timeline Summary

Week	Goal	Status
1-2	Terminal chatbot working	DONE
3	Voice input + output	Next
4-5	Memory across sessions	Pending
6-8	Actions + intent detection	Pending
9-10	GitHub + demo + resume	Pending

Professional Identity Notes

Recommended professional email: `sanjay.k.robotics@gmail.com` — clean, field-specific, resume-ready.
Current email `sanjaypro794@gmail.com` is not suitable for professional use.

Long-term domain option: `sanjayk-robotics.dev` (~Rs.700/year on Namecheap). Email becomes `sanjay@sanjayk-robotics.dev` — strong positioning for Siemens, ABB, Honeywell applications.